

# ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Models

Email: [atu1@umd.edu](mailto:atu1@umd.edu)

LinkedIn: [linkedin.com/in/allentu](https://www.linkedin.com/in/allentu)

Website: [tuallen.github.io](https://tuallen.github.io)

Google Scholar: [Allen Tu](#)

## SELECTED PUBLICATIONS

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1. **Allen Tu\***, H. Ying\*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *CVPR*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *CVPR*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *FG*, 2026.
4. A. Hanson\*, **Allen Tu\***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’. *CVPR*, 2025.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’. *CVPR*, 2025.

\* denotes equal contribution.

## EDUCATION

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**University of Maryland, College Park**

Ph.D. Student, Computer Science

College Park, MD

January 2025 – May 2028 (Expected)

- Advised by Professor Tom Goldstein; collaborating with Professor Matthias Zwicker

B.S./M.S. in Computer Science, Minor in Statistics

August 2019 – December 2024

## RESEARCH EXPERIENCE

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**Amazon**

Applied Science Intern

Bellevue, WA

June 2026 – October 2026 (Expected)

- Incoming Applied Science Intern on the Last Hundred Yards team, focusing on 3D reconstruction for robotics

**University of Maryland Institute of Advanced Computer Studies**

Graduate Research Assistant

College Park, MD

August 2023 – Present

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude ([WRIVA](#)) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over 10× fewer primitives while preserving visual fidelity [1, 4, 5]
- Introduced uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [2]

**Systems & Technology Research**

Computer Vision Research Intern

Arlington, VA

May 2025 – January 2026

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range ([BRIAR](#)) program
- Proposed an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at 1e-3 FMR using an encoder trained solely on public data [3]

## Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

May 2024 – August 2024

- Achieved 0.86 correlation with ground-truth probe-to-gallery similarity and state-of-the-art ERC performance for face image quality estimation (FIQA), outperforming prior methods on image-level evaluation [3]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 12%

Computer Vision Research Co-op

January 2023 – August 2023

- Implemented a mixed-voting ensemble that improved open-set search performance by 24% for face recognition, 39% for body recognition, and 49% for fused multimodal biometric recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52% in severe operating conditions

Computer Vision Research Intern

June 2022 – August 2022

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

## Undergraduate Researcher

University of Maryland Department of Computer Science

College Park, MD

January 2021 – December 2022

- Analyzed 2.3 million dataset-visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [6]

## RESEARCH PROJECTS

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### IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)

University of Maryland Institute of Advanced Computer Studies

August 2023 – Present

Unconstrained 3D reconstruction and novel view synthesis in challenging real-world environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [1, 4, 5]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [2]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [4]

### IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)

Systems & Technology Research

June 2022 – January 2026

Multimodal, opportunistic fusion of incomplete face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment [3]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling
- Style-Based Appearance Flow for Clothing-Robust Body Representation Learning

## SERVICE

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### SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models

Workshop Co-organizer

CVPR 2026

- Co-organized the first CVPR workshop on security, privacy, and robustness in 3D generative vision models, coordinating peer review, securing \$1K in sponsorship, and leading industry outreach

**Professional Memberships:** IEEE Member; Computer Vision Foundation (CVF); IEEE Biometrics Council

## TEACHING EXPERIENCE

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### Peer Research Mentor

College Park, MD

*FIRE: Capital One Machine Learning*

*January 2021 – December 2022*

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition

## TECHNICAL SKILLS

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**Languages:** Python, CUDA, C/C++, C#, Java, SQL, Shell

**Frameworks & Tools:** PyTorch, TensorFlow, NumPy, SciPy, OpenCV, scikit-learn

**Research Areas:** 3D/4D Reconstruction, 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Biometrics, Diffusion Models, Vision Transformers (ViT/Swin), Uncertainty Quantification, Scalable Vision Systems, GPU Programming